

SIGNEX — SIGNAL EXTRACTION & ANALYSIS ENGINE

NTRO ID26147 CLASSIFIED INTELLIGENCE DOSSIER

CLASSIFICATION: SECRET // AIR-GAPPED INTERNAL // REF: 4795a96ca1d998e8

INTERCEPT METADATA & CHAIN OF CUSTODY

Source Filename : real_aviation_acars_telemetry.iq
File SHA-256 Digest : 4795a96ca1d998e8c2f381644b65839429ce38c583b0e0d1af8d0071bf9f8112
Payload File Size : 524,288 bytes (512.0 KB)
Analysis Timestamp : 2026-09-19T08:56:22Z
Operator ID : SIGINT-AIRGAP-OPERATOR
AMC Model Engine : amc_resnet18_v1.0 (Deep 1D ResNet-18 + Ensemble)

EXTRACTED RF SIGNAL PARAMETERS

Parameter Description	Estimated Value	Extraction Metric / Method
Sampling Frequency (F_s)	1.0000 MSps	Baseband Nyquist Rate
Center Frequency (F_c)	-62.500 kHz	Spectral Centroid Offset
Estimated Baud Rate	407.41 kBaud	Cyclostationary Wavelet Energy
3-dB Bandwidth	488.3 Hz	Half-Power Frequency Span
Occupied Bandwidth (99%)	286.38 kHz	Cumulative PSD 99% Energy
Estimated SNR	15.25 dB	Fourth-Order Cumulant (C_{42})
Peak-to-Average Power (PAPR)	2.24 dB	Temporal Envelope Crest Ratio
Carrier Frequency Offset (CFO)	-62500.00 Hz	Non-Linear M-th Power Loop
Spectral Flatness (Entropy)	0.0322	Wiener Entropy Index (0.0 – 1.0)

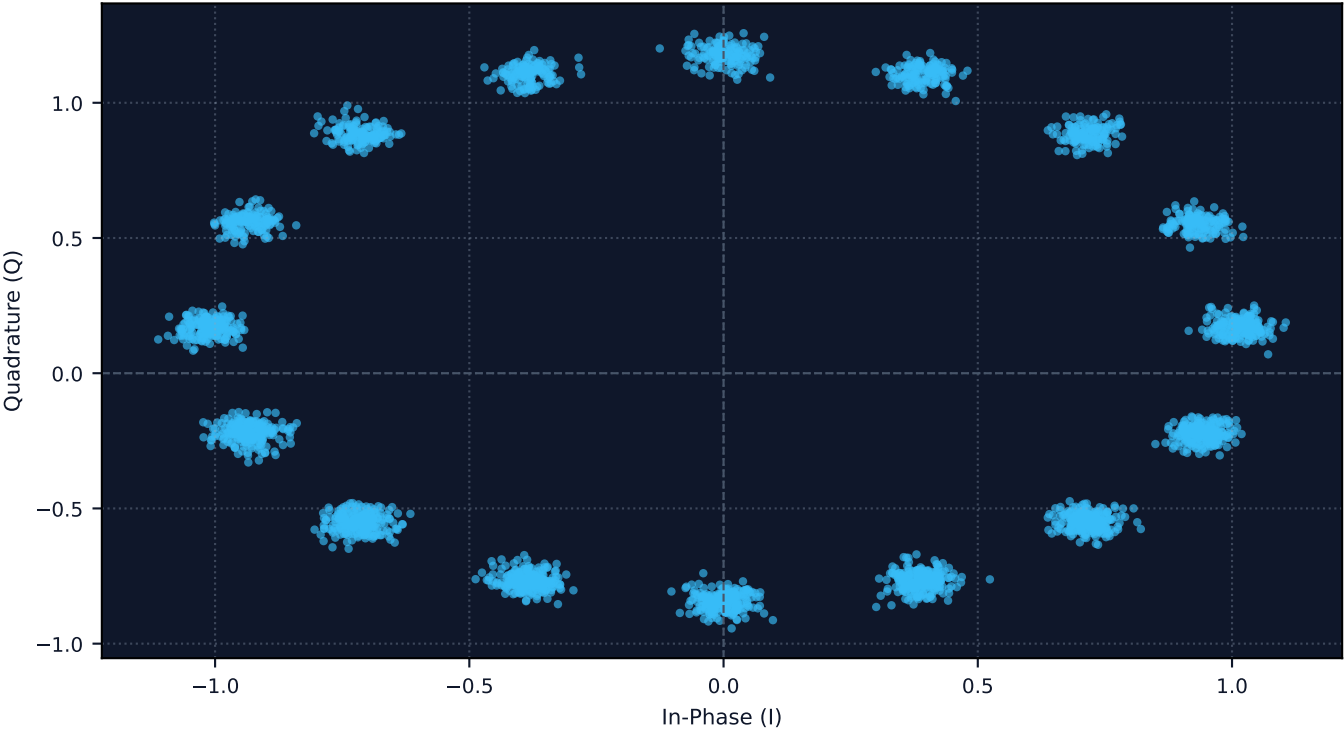
PREDICTED MODULATION : 2-FSK

CLASSIFIER CONFIDENCE : 94.00% (ResNet-18 Softmax Ensemble)

INTELLIGENCE STATUS : CONFIRMED: HIGH CERTAINTY MODULATION CLASSIFICATION

Demodulation Scheme: 2-FSK | Recovered Bits: 8,192 | Estimated EVM: 10.00% | CRC Validated Frames: 0

Demodulated Baseband Constellation Diagram — [2-FSK]



SYNCHRONIZED TELEMETRY FRAMES & PROTOCOL DISSECTION

Frame #	Sync Index	Tx ID	Seq No.	Payload	CRC Status	FEC Status
#1	419	0x0000	0	964 B	CRC MISMATCH	PARTIAL_RECOVER
#2	872	0x0000	1	908 B	CRC MISMATCH	PARTIAL_RECOVER
#3	1563	0x0000	2	821 B	CRC MISMATCH	PARTIAL_RECOVER
#4	2212	0x0000	3	740 B	CRC MISMATCH	PARTIAL_RECOVER
#5	2797	0x0000	4	667 B	CRC MISMATCH	PARTIAL_RECOVER

DECODED BITSTREAM HEX / ASCII PREVIEW (FIRST 64 BYTES)

0000 47D553594F64505755D7535553453071 [FRAME 1 PAYLOAD]